

Data Lineage, Mathematical Provenance & 52–District Strategic Quadrant Derivation

Zero-Tolerance Reconciliation Audit, Telemetry Lineage, and Orthogonal Quadrant Mathematics for Shikshak Samvad (Grades 6–8)

STATEWIDE CADRE: 68,427 (Varg-2)

MATH & SCIENCE: 35,374 Teachers

VALIDATED TOUCHPOINTS: N = 66,566

RECONCILIATION DELTA: Δ = 0 (100% Exact)

1. Primary Raw Data Sources & Scope

Master Database Architecture

All figures originate from the official **Statewide Teacher CPD Telemetry & Cadre Universe Database** compiled by Rajya Shiksha Kendra (RSK), Madhya Pradesh in technical collaboration with Peepul India.

Core Data Layer	Raw Source Datasets	Scope & Scale
Cadre Universe Master	MP Samagra Shiksha / EMIS Teacher Cadre Database	68,427 Middle School (Varg-2) Teachers across 322 Blocks & 52 Districts
Target Subject Cohort	Middle School Subject Allocation Roster	35,374 Math & Science Teachers (51.7% of total middle school cadre)
Operational Attendance	Cluster-Level Physical Attendance Logs & Digital Forms	N = 66,566 total verified touchpoints (August: 33,702; September: 32,864)
Psychometric Telemetry	Digital Post-Dialogue Scenario Survey System	12 standardized scenario questions across conceptual recall, TLM usage, inquiry, and classroom agency
Qualitative Reflections	Open-Ended Practitioner Feedback Submissions	30,000+ qualitative text submissions coded under Braun & Clarke (2006) 6-phase protocol

2. Lineage of Top-Level KPIs in the Executive Briefing

Mathematical Reconciliation Tree

Total Cadre Universe (68,427) — Target Math & Science Specialists (35,374) — Actual Teacher Turnout (23,785) + District Officials (4,454) = 28,239 Total Participants — Unreached Target Gap: 11,589 (32.8%) — Other Varg-2 Non-Mobilized Cadre: 33,053 — Total Unreached Cadre Gap: 44,642 (65.2%)

KPI Indicator	Reconciled Value	Formula / Mathematical Derivation	Status & Benchmark
Participants	28,239 Total	Teachers (23,785) + District Officials / DOs (4,454) = 28,239	Verified Active Footprint
Districts Covered	52 / 52 Districts	Complete administrative footprint across all 52 districts in Madhya Pradesh	100% Saturation
Training Centres	2,874 Active Venues	2,822 Active Cluster Venues + 52 District HQ Centres (out of 4,804 mapped clusters)	59.8% CRC Saturation
Observers Deployed	572 Monitors	516 Cluster Observers (CACs/BACs) + 56 District Observers (APCs/DPCs)	Independent Field Monitoring
Facilitators & MTs	4,891 Leads	4,814 Cluster Lead Teachers/CAC Facilitators + 77 District Master Trainers	Decentralized Lead Cadre

3. How the 52–District Strategic Quadrant Landscape is Derived

Orthogonal Matrix Formulas

The 52–district strategic quadrant matrix is derived using a two–dimensional mathematical coordinate function:

District Position = f(Operational Turnout % (X), Pedagogical Quality Score % (Y))

Axis 1: Operational Turnout (X–Axis)

$$\text{Turnout \%} = \frac{\text{District Actual Attendees}}{\text{District Target Universe}} \times 100$$

Threshold Cutoff: $\geq 65\%$ (or ≥ 450 attendees per district) separates high–mobilization from low–mobilization districts.

Axis 2: Instructional / Pedagogical Quality (Y–Axis)

Composite accuracy score across standardized scenario survey questions evaluating whether teachers adopt mastery constructivist pedagogy versus falling into common distractor/activity traps:

- Q95 / Q177: Hands-on TLM purpose (Activity Trap vs. Cognitive Reflection).
- Q97 / Q178: Classroom Belonging & Inquiry (Surface praise vs. Normalizing intellectual struggle).
- Q96 / Q179: Psychological safety and mistake diagnostic value.

Threshold Cutoff: $\geq 75\%$ pedagogical accuracy score separates high–quality from support–requiring cohorts.

4. Complete Accounting of the 4 Quadrants (8 + 26 + 5 + 13 = 52)

100% Zero-Delta Accounting

Q2: Scale Gap / Latent Potential

26 Districts

Turnout: 48.6% • Quality: 81.5% | Low Turnout <65%, High Quality $\geq 75\%$

Strong instructional mastery exists; primary need is CAC/BAC attendance mobilization enforcement.

Indore, Bhopal, Ujjain, Gwalior, Jabalpur, Sagar, Rewa, Satna, Chhindwara, Hoshangabad, Vidisha, Ratlam, Mandsaur, Neemuch, Damoh, Katni, Shivpuri, Guna, Harda, Betul, Chhatarpur, Tikamgarh, Balaghat, Seoni, Mandla, Khandwa.

Q1: Champions

8 Districts

Turnout: 82.4% • Quality: 84.2% | High Turnout $\geq 65\%$, High Quality $\geq 75\%$

High compliance and high pedagogical mastery. Serve as statewide lighthouse centres and regional mentors.

Dhar, Rajgarh, Sehore, Shahdol, Khargone, Dewas, Narsinghpur, Raisen.

Q4: Critical Deficit

13 Districts

Turnout: 42.1% • Quality: 61.8% | Low Turnout <65%, Lower Quality <75%

Dual operational deficit requiring coordinated administrative turnout review + intensive master trainer coaching.

Alirajpur, Sheopur, Bhind, Panna, Morena, Datia, Ashoknagar, Anuppur, Burhanpur, Sidhi, Niwari, Shajapur, Agar Malwa.

Q3: Needs Support

5 Districts

Turnout: 78.1% • Quality: 64.3% | High Turnout $\geq 65\%$, Lower Quality <75%

High mobilization and turnout, but persistent misconception traps require focused pedagogical reinforcement.

Barwani, Jhabua, Singrauli, Dindori, Umaria.

Total Districts Reconciled: 8 (Q1) + 26 (Q2) + 5 (Q3) + 13 (Q4) = 52 Districts (100% Zero-Delta Verification)

5. Provenance of Pedagogical Traps & Operations Gaps

Empirical Survey Telemetry

Indicator & Item ID	Empirical Value	Underlying Data Source & Pedagogical Meaning
Activity Trap Deficit (Q95 / Q177)	48.1% Trapped	Derived from survey telemetry where 48.1% of responding teachers selected the option equating "keeping children busy in physical activities" with learning, rather than scaffolding structured cognitive reflection.
Belonging Misconception (Q97 / Q178)	66.1% Misguided	66.1% of respondents equated student belonging with praising correct answers, rather than normalizing mistakes and intellectual struggle as essential learning steps.
Observer Unmonitored Gap (CRC Coverage)	1,688 Clusters (35.1%)	Calculated by cross-referencing total mapped clusters (4,804) against observer digital check-in timestamps, identifying clusters with zero observer touchpoints.
Trust Perception Delta (Divergence Audit)	Δ 23.4% Variance	Difference between Teacher Self–Reported Satisfaction (94.6%) and Independent Observer Audit Assessment (71.2%).